Controls for the Jacobian Lens

Morgan Hough

2026-07-19

Independent

Introduction

The claim

July 2026. Anthropic: *"Verbalizable Representations Form a Global Workspace in Language Models."* Claude contains a "J-space" functionally analogous to the global workspace of Baars and Dehaene. 1.2M views in a day.

Reported properties: reportability, controllability, causal role in reasoning, flexible reuse — and **limited scope**: ablating the J-space destroys multi-step reasoning and summarisation while sparing fluency, sentiment, and factual recall.

The method. A Jacobian lens:

$$\mathrm{lens}_l(h) = \mathrm{unembed}(J_l\, h), \qquad J_l = \mathbb{E}\left[\partial h_{\mathrm{final}} / \partial h_l\right]$$

Apache-2.0, plus 38 fitted lenses for open models.

The public response

Two camps. **Neither ran the code.**

(1) “It’s just backprop.” *(A. Trask)* — but that is the wrong derivative. Backprop takes $\partial L/\partial\theta$: needs a label, moves weights. The J-lens takes $\partial h_{\mathrm{final}}/\partial h_l$ on a *frozen* model. *The steelman is better:* J_l is an averaged linear map into the final residual basis — an analytically-derived **tuned lens**.

(2) “Unfalsifiable by construction.” *(E. Hoel)* — strip global workspace theory to reportability and it becomes trivial; predictions and inferences “can never truly be pulled apart.” His one empirical falsification rests on a third-party demo he declines to stand behind: a real replication “would be important to see.”

State of the art for adjudicating a consciousness claim: two rhetorical positions and no controls.

Background

One move, four papers

Write a direction into the residual stream. Read an effect. Name it after a construct from cognitive science.

	Direction from	**Readout**	**Construct**
Manifolds (Oct 25)	PCA of condition means	activation space	place cells
Introspection (Oct 25)	difference-in-means	verbal report	metacognition
Societies (Jan 26)	SAE feature	task accuracy	society of thought
J-space (Jul 26)	Jacobian lens	**vocabulary**	global workspace

Lineage is explicit: J-space reuses the manifolds counting task *and* Lindsey's injection protocol. Shared authors throughout; Lindsey is final author.

The contribution is the readout space. PCA components have no words attached; a vocabulary-indexed readout is what makes "verbalizable" a meaningful predicate.

What each paper controls for

Control	Manif.	Intro.	SoT	J-space
matched random subspace	**Y**	—	—	—
norm-matched random vector	—	**Y**	—	—
negative / inverted direction	—	**fails**	—	—
no-intervention baseline	—	**Y**	**Y**	**Y**
model randomization	—	—	—	—
drift null (geometry)	—	n/a	n/a	—

Lindsey's introspection paper is the best controlled of the four — and its weakness is one it reports itself: injecting the *negation* of a concept vector was "comparably effective," words showing "no discernible pattern."

No paper runs a model-randomization control (Adebayo et al. 2018). Notably, a co-author of *An Interpretability Illusion for BERT* is on two of these papers — the expertise is in the room.

Results

Result 1 — the lens passes randomization

Randomize the transformer *blocks*; keep the **trained** embedding, final norm and unembedding — so token identity stays meaningful and the control cannot pass for the wrong reason. Score two ways per layer.

	peak next_acc	**peak echo**
trained	0.3414	0.2936
random blocks	**0.0003**	**0.0016**

Trained shows a clean crossover: early layers echo the current token, then echo decays as next-token prediction climbs. **Random blocks read out nothing.**

The J-lens passes. Its structure requires learned weights. The strong form of "it's just the residual stream" is refuted — and Anthropic never ran this control. *Controls should be able to exonerate as well as convict.*

Result 2 — the metric rewards noise

The paper's lens-quality score is **minimum rank over ∼35 layers**. That hands a diffuse readout *one lottery ticket per layer*. A confident lens correctly reading the content puts an unrelated word far down at *every* layer.

Qwen3.5-27B, at the ^ of the ASCII face, on **Anthropic's own lens**:

lens	**rank("nose")**	**top-5**
J-lens	**2**	`smile, nose, '^, noses, grin`
logit lens	5	`\n, ~, .., -, N`

Comparable scores. One has understood it is looking at a face; the other emits punctuation. **`pass@k` cannot tell them apart.**

This is why the logit lens "beat" the J-lens on mean `pass@10` at 4B and 8B in our first pass — an artifact of the metric, which we nearly reported as a refutation.

Result 3 — the tripartite figure is mostly drift

The sensory/workspace/motor block structure is the one part Hoel concedes is *not* baked in by construction. Build a **distance-only null**: entries depending solely on $|i - j|$, same decay profile, *zero blocks by construction*.

model	real	null	excess
qwen3-1.7b	0.117	0.092	+0.025
qwen3-4b	0.078	0.062	+0.016
qwen3-8b	0.283	0.249	+0.033
qwen3-14b	0.293	0.267	+0.026
gpt-oss-20b	0.210	0.157	**+0.053**
qwen3.5-27b	0.267	0.217	**+0.050**

The null recovers 79–91%. Excess doubles at $\geq$20B but stays small. Raw blockiness rising 0.08$\rightarrow$0.29 with scale is the *decay profile steepening*, not blocks appearing.

Both sides overclaimed: Anthropic on the sharpness, Hoel on the absence — his substitution argument generalises from small models, the regime where there is nothing

Result 4 — two retractions

I reported a sharp emergence threshold. The ASCII-face readout surfaced at Qwen3.5-27B (rank 2) and not at dense Qwen3-14B (rank 164). Then I reported a *correction*: it tracked architecture, not scale.

Both were wrong. The robust version — the 102-vignette `association` eval, concepts evoked but never named — gives a smooth monotone rise in which a **dense** 32B beats the 27B hybrid at every rank:

model	**J@1**	**J@10**	**J@50**
14B dense	0.040	0.091	0.222
27B hybrid	0.049	0.167	0.343
32B dense	**0.062**	**0.208**	**0.438**

The original finding was **one lucky prompt**. Architecture-dependence is *untested*, not established

Result 5 — post-training moves the J-space, but the tidy reading is confounded

Claim 6: post-training shaped the J-space *"toward a point of view"* rather than pure prediction. Untestable from outside — until OLMo-3. It is the only open family shipping a base model, its post-trained variants, the **public training data** (Dolma), and ∼1,486 checkpoints.

Magnitude — holds. $\mathrm{mean}\ \cos(J_{\text{base}}, J_{\text{instruct}}) = 0.76$ over 8 pairs, against a ∼0.96 same-model refit-noise floor. Post-training *does* move the workspace, well beyond fitting noise. First outside quantification of the claim.

Structure — confounded. The tempting next step: $dJ = J_{\text{instruct}} - J_{\text{base}}$ looks strongly low-rank (rank fraction 0.003–0.02 vs 0.50 for iid drift) — a "structured viewpoint shift." But J is *already* low-rank, and the change inherits it:

model	**rank**(J_{base})	**rank**(dJ)	**ratio**
gemma-3-270m	0.048	0.022	0.45x
gemma-3-1b	[illegible]	[illegible]	[illegible]

Result 6 — one model's workspace holds a suppressed register

A cheap, no-GPU content probe (divide the embedding norm out through the motor band) suggested OLMo-3-7B's workspace holds discourse markers over literal next-tokens. $n=1$, hand-picked layers. **So I pre-registered a gate** and ran it band-wide, with a 1000-permutation null, across 7 cross-family models.

As a general claim it fails: significant in 1 of 7. I had over-generalised an OLMo-specific result by calling it "the workspace." The general line is dropped.

But OLMo was an 11.6$\times$ outlier among $\sim$1$\times$, and it is real and broader than discourse. Its workspace band preferentially expresses an informal / negative / profane / connective register its *output* layers suppress:

lexicon	**OLMo-3-7B**		**Qwen3-8B (same tokens+measure)**	
discourse	11.5x	$p=0$	1.3x	n.s.
negative	12.1x	$p=0$	0.0x	n.s.
profanity	8.6x	$p=0$	4.1x	marg.

Result 7 — Claim 6, tested on fully open weights

Anthropic's Claim 6: post-training shaped the J-space *"toward a point of view rather than pure prediction."* It rests on Sonnet 4.5 — whose activations no outsider can touch; the paper's own commentators (Dehaene & Naccache) could not check it. **OLMo-3 makes it checkable on FULLY OPEN artifacts.** First external test.

Post-training moves the J-space — a lot, and by METHOD not domain. Cosine of each arm's lens against the base lens, vs a 0.97 same-model refit floor:

arm	cos(base, arm)	move
Instruct (SFT+DPO)	0.69	∼31**%**
Think (SFT+DPO)	0.73	∼27%
RL-Zero (RLVR only)	0.94	∼6%
RL-Zero domain pairwise	**0.99+**	∼1% (Math/Code/IF/General)

Instruction/CoT tuning reshapes the J-space **about 5x more than RLVR**, and varying the RLVR *domain* at matched capability adds only about 1%. The viewpoint is

Discussion

Who is allowed to check any of this?

No external party holds activations for Claude, GPT or Gemini. There is no programme to apply to. The published ceiling (METR, May 2026) is black-box plus raw chain-of-thought, under NDA, with the lab approving publications — and Apollo and UK AISI got **under a week** with Sonnet 4.5.

How did the invited commentators check the J-space claims? **They didn't.** Nanda received "an advance draft" and replicated on **Qwen3.5-27B**. Dehaene and Naccache: *"We suggested to the Anthropic team that they could run exactly the same tests..."* — the originators of the theory could not test it themselves.

And the checkability is borrowed. Neuronpedia — an MIT-licensed project created June 2023 and run by *one person* — fits and hosts these lenses, alongside DeepMind's and OpenMOSS's SAEs. Anthropic's own HuggingFace org publishes zero models. The convergence criterion that is missing from the public chain lives in *Neuronpedia's* unreleased wrapper.

Conclusion

Where this leaves the claim

What survives. The J-lens is a real instrument: it passes randomization, and it reads content at mid-layers that a logit lens cannot. The ablation dissociation — fluency and recall spared, multi-step reasoning destroyed — is a genuine finding.

What does not. The sharp tripartite geometry is mostly smooth drift. The lens-quality metric rewards noise. The emergence story was one prompt. The "structured point-of-view shift" from post-training is a low rank dJ inherits from J. And the paper itself concedes the architecture: *"no obviously separable input processors"*; broadcast "within a single feedforward pass rather than through recurrent loops."

Two positive results, both on fully open weights. (1) On the OLMo-3 ladder, post-training reshapes the J-space ∼31% (Instruct) while capability stays flat — a viewpoint shift decoupled from prediction, method-driven and domain-invariant. Claim 6 supported on artifacts anyone can rerun. (2) OLMo-3-7B's workspace holds a

Questions?